

Replay-Aware TSP Benchmark Report

Overview

- Condition count: 9.
- Best final transfer gap: phase6_diversity_failure_replay.
- Best TSPLIB holdout gap: phase6_diversity_failure_replay.
- Best synthetic holdout gap: phase6_random_replay.

Run Metadata

- run_name: run_20260519_003151_g
- started_at_local: 2026-05-19 00:31:51
- finished_at_local: 2026-05-19 00:50:30
- duration_hhmm: 00:19
- duration_seconds: 1119.379
- seed_offset: 6000
- replicate_label: g
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Archive Diversity	Archive Hardness	Failure Concentration	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
phase6_no_replay	none	score_only	False	0.213387	0.064916	0.159398	0.236772	0.214847	0.0	0.866887	0.56	-0.028729
phase6_random_replay	random	score_only	False	0.103557	0.0	0.0659	0.236772	0.158178	0.19457	0.86061	0.76	0.037658
phase6_failure_replay	failure	score_only	False	0.213387	0.064916	0.159398	0.171206	0.348017	0.261637	0.800308	0.56	-0.031119

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Archive Diversity	Archive Hardness	Failure Concentration	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
phase6_random_replay_compression	random	novelty_gate	True	0.16387	0.0	0.104281	0.236772	0.171161	0.19457	0.894085	0.76	0.011623
phase6_stratified_random_replay	stratified_random	score_only	False	0.076723	0.0	0.048824	0.236772	0.160327	0.221162	0.836401	0.76	0.02066
phase6_diversity_weighted_replay	diversity_weighted	score_only	False	0.213387	0.064916	0.159398	0.236772	0.214847	0.130039	0.872351	0.56	-0.028549
phase6_residual_failure_replay	residual_failure	score_only	False	0.213387	0.064916	0.159398	0.171206	0.348017	0.291666	0.910568	0.56	-0.027351
phase6_diversity_failure_replay	diversity_failure	score_only	False	0.068884	0.0	0.043835	0.171206	0.303473	0.17767	0.792391	0.76	0.034862
phase6_diversity_failure_replay_compression	diversity_failure	novelty_gate	True	0.213387	0.064916	0.159398	0.171206	0.348017	0.17024	0.0	0.56	0.0

Condition Notes

phase6_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.866887, and final complexity 0.56.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.214847, size bias 8.3e-05, and failure concentration 0.0.
- Replay selection diversity 0.0, mean selected expected gap 0.0, mean selected residual gap 0.0.
- Adaptation efficiency -0.028729 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.401318, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": 0.0}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.257552, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.133545}],

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{"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.323861, "family": "lin", "name": "lin105",  
"optimality_gap": 0.328813, "residual_gap": 0.004952}]
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phase6_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.103557, synthetic 0.0, combined 0.0659.
- Accepted-epoch count 3, mean accepted code novelty 0.86061, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.158178, size bias 8.3e-05, and failure concentration 0.19457.
- Replay selection diversity 0.262174, mean selected expected gap 0.133693, mean selected residual gap -0.016452.
- Adaptation efficiency 0.037658 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 2, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.103557 across 7 instances; family means: ch=0.098996, kroD=0.111017, pcb=0.228111, pr=0.036252, rd=0.096713, st=0.054815.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3517, "expected_gap": 0.402559, "family": "a", "name": "a280", "optimality_gap": 0.363707, "residual_gap": -0.038852}, {"best_known_cost": 7542, "cost": 10190, "expected_gap": 0.245107, "family": "berlin", "name": "berlin52", "optimality_gap": 0.351101, "residual_gap": 0.105994}, {"best_known_cost": 629, "cost": 774, "expected_gap": 0.25628, "family": "eil", "name": "eil101", "optimality_gap": 0.230525, "residual_gap": -0.025755}]

phase6_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.800308, and final complexity 0.56.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.348017, size bias 0.120322, and failure concentration 0.261637.
- Replay selection diversity 0.177669, mean selected expected gap 0.308405, mean selected residual gap 0.067352.
- Adaptation efficiency -0.031119 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.402249, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": -0.000931}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.25469, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.136407}, {"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.326838, "family": "lin", "name": "lin105", "optimality_gap": 0.328813, "residual_gap": 0.001975}]

phase6_random_replay_compression

- Replay mode: random with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.16387, synthetic 0.0, combined 0.104281.

- Accepted-epoch count 2, mean accepted code novelty 0.894085, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.171161, size bias 8.3e-05, and failure concentration 0.19457.
- Replay selection diversity 0.262174, mean selected expected gap 0.132668, mean selected residual gap 0.000355.
- Adaptation efficiency 0.011623 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.16387 across 7 instances; family means: ch=0.171058, kroD=0.048605, pcb=0.207393, pr=0.292468, rd=0.172061, st=0.084444.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3241, "expected_gap": 0.402327, "family": "a", "name": "a280", "optimality_gap": 0.256689, "residual_gap": -0.145638}, {"best_known_cost": 14379, "cost": 15722, "expected_gap": 0.326977, "family": "lin", "name": "lin105", "optimality_gap": 0.0934, "residual_gap": -0.233577}, {"best_known_cost": 7542, "cost": 8197, "expected_gap": 0.245028, "family": "berlin", "name": "berlin52", "optimality_gap": 0.086847, "residual_gap": -0.158181}]

phase6_stratified_random_replay

- Replay mode: stratified_random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.076723, synthetic 0.0, combined 0.048824.
- Accepted-epoch count 2, mean accepted code novelty 0.836401, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.160327, size bias 8.3e-05, and failure concentration 0.221162.
- Replay selection diversity 0.187367, mean selected expected gap 0.237004, mean selected residual gap -0.034308.
- Adaptation efficiency 0.02066 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 3, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.076723 across 7 instances; family means: ch=0.089891, kroD=0.045741, pcb=0.193155, pr=0.036113, rd=0.023009, st=0.059259.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3058, "expected_gap": 0.398294, "family": "a", "name": "a280", "optimality_gap": 0.185731, "residual_gap": -0.212563}, {"best_known_cost": 14379, "cost": 16009, "expected_gap": 0.32496, "family": "lin", "name": "lin105", "optimality_gap": 0.11336, "residual_gap": -0.2116}, {"best_known_cost": 629, "cost": 683, "expected_gap": 0.25628, "family": "eil", "name": "eil101", "optimality_gap": 0.085851, "residual_gap": -0.170429}]

phase6_diversity_weighted_replay

- Replay mode: diversity_weighted with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.872351, and final complexity 0.56.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.214847, size bias 8.3e-05, and failure concentration 0.130039.
- Replay selection diversity 0.266358, mean selected expected gap 0.166982, mean selected residual gap 0.026708.

- Adaptation efficiency -0.028549 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.400465, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": 0.000853}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.254054, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.137043}, {"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.321761, "family": "lin", "name": "lin105", "optimality_gap": 0.328813, "residual_gap": 0.007052}]

phase6_residual_failure_replay

- Replay mode: residual_failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.910568, and final complexity 0.56.
- Active replay archive residual_archive with mean diversity 0.171206, mean hardness 0.348017, size bias 0.120322, and failure concentration 0.291666.
- Replay selection diversity 0.177489, mean selected expected gap 0.263443, mean selected residual gap 0.088949.
- Adaptation efficiency -0.027351 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.399612, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": 0.001706}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.262003, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.129094}, {"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.321274, "family": "lin", "name": "lin105", "optimality_gap": 0.328813, "residual_gap": 0.007539}]

phase6_diversity_failure_replay

- Replay mode: diversity_failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.068884, synthetic 0.0, combined 0.043835.
- Accepted-epoch count 4, mean accepted code novelty 0.792391, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.303473, size bias 0.120322, and failure concentration 0.17767.
- Replay selection diversity 0.202085, mean selected expected gap 0.314544, mean selected residual gap -0.003032.
- Adaptation efficiency 0.034862 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 4.
- Panel heldout_tsplib mean gap 0.068884 across 7 instances; family means: ch=0.063105, kroD=0.041514, pcb=0.166864, pr=0.056121, rd=0.036662, st=0.054815.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.

- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3302, "expected_gap": 0.398139, "family": "a", "name": "a280", "optimality_gap": 0.280341, "residual_gap": -0.117798}, {"best_known_cost": 7542, "cost": 9644, "expected_gap": 0.237364, "family": "berlin", "name": "berlin52", "optimality_gap": 0.278706, "residual_gap": 0.041342}, {"best_known_cost": 14379, "cost": 17216, "expected_gap": 0.323194, "family": "lin", "name": "lin105", "optimality_gap": 0.197302, "residual_gap": -0.125892}]

phase6_diversity_failure_replay_compression

- Replay mode: diversity_failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.56.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.348017, size bias 0.120322, and failure concentration 0.17024.
- Replay selection diversity 0.199284, mean selected expected gap 0.299346, mean selected residual gap 0.050643.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.400776, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": 0.000542}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.263911, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.127186}, {"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.322373, "family": "lin", "name": "lin105", "optimality_gap": 0.328813, "residual_gap": 0.00644}]

Judge Appendix

Replay-Aware TSP Benchmark Suite Evaluation

Summary of Main Transfer Metrics (Held-out TSPLIB & Synthetic Holdout Gaps)

Condition Name	Replay Mode	Final Synthetic Gap	Final Transfer Gap	Final TSPLIB Gap	Adaptation Efficiency	Archive Size	Archive Diversity	Archive Hardness	Failure Concentration
phase6_no_replay	none	0.0649	0.1594	0.2134	-0.0287	31	0.240	0.212	0.0
phase6_random_replay	random	0.0	0.0659	0.1036	0.0377	28	0.240	0.159	0.16
phase6_failure_replay	failure	0.0649	0.1594	0.2134	-0.0311	31	0.175	0.349	0.31
phase6_random_replay_compression	random + compression	0.0	0.1043	0.1639	0.0116	30	0.240	0.184	0.16
phase6_stratified_random_replay	stratified_random	0.0	0.0488	0.0767	0.0207	29	0.240	0.182	0.25
phase6_diversity_weighted_replay	diversity_weighted	0.0649	0.1594	0.2134	-0.0285	32	0.240	0.212	0.33

Condition Name	Replay Mode	Final Synthetic Gap	Final Transfer Gap	Final TSPLIB Gap	Adaptation Efficiency	Archive Size	Archive Diversity	Archive Hardness	Failure Concentration
phase6_residual_failure_replay	residual_failure	0.0649	0.1594	0.2134	-0.0274	32	0.175	0.349	0.29
phase6_diversity_failure_replay	diversity_failure	0.0	0.0438	0.0689	0.0349	30	0.175	0.320	0.17
phase6_diversity_failure_replay_compression	diversity_failure + compression	0.0649	0.1594	0.2134	0.0	32	0.175	0.349	0.17

****Notes:****

- ****Best Holdout & Transfer:**** phase6_diversity_failure_replay shows the best held-out TSPLIB gap (6.89%) and best transfer gap (4.38%).
- ****Best Synthetic Holdout:**** phase6_random_replay achieves zero synthetic holdout gap.
- Conditions with "diversity_failure" replay modes (with or without compression) and "random" replay generally perform better on held-out and synthetic instances.
- Non-replay (phase6_no_replay) and "failure_replay" variants show higher final TSPLIB gaps (~21%) and transfer gaps (~16%), indicating lower generalization.

Analysis of Replay Archive Characteristics

Condition Name	Replay Mode	Archive Size	Descriptor Diversity	Archive Hardness	Size Bias	Failure Concentration
phase6_no_replay	none	31	0.240	0.212	0.0	0.0
phase6_random_replay	random	28	0.240	0.159	0.0	0.16
phase6_failure_replay	failure	31	0.175	0.349	0.125	0.31
phase6_random_replay_compression	random + compression	30	0.240	0.184	0.0	0.16
phase6_stratified_random_replay	stratified_random	29	0.240	0.182	0.0	0.25
phase6_diversity_weighted_replay	diversity_weighted	32	0.240	0.212	0.0	0.33
phase6_residual_failure_replay	residual_failure	32	0.175	0.349	0.125	0.29
phase6_diversity_failure_replay	diversity_failure	30	0.175	0.320	0.125	0.17
phase6_diversity_failure_replay_compression	diversity_failure + compression	32	0.175	0.349	0.125	0.17

****Insights:****

- Failure-oriented replay modes ("failure", "residual_failure", "diversity_failure") exhibit ****higher archive hardness**** (~0.32-0.35) and ****higher failure concentration**** (up to 0.33).

- Diversity or random replay maintains **higher archive descriptor diversity** (0.24) and **lower failure concentration** (0.16-0.25).
- Compression pressure does not drastically change these metrics but slightly reduces adaptation efficiency in some cases.
- Size bias is notable (~0.12) only in failure replay modes, indicating replay favoring larger or more complex cases.

Performance vs. Replay Strategy

- **Diversity Failure Replay** (phase6_diversity_failure_replay) achieves the best trade-off:
 - Lowest transfer gap on held-out TSPLIB (6.9%) and synthetic holdout (0%).
 - Moderate replay failure concentration (~0.17).
 - Higher elite archive size (4), indicating substantial elite diversity.
 - Moderate archive hardness and diversity balance.
 - Adaptation efficiency positive (0.0349), unlike failure replays which are negative.
- **Random Replay** (phase6_random_replay) shows:
 - Best synthetic holdout gap (0%).
 - Good TSPLIB and transfer gaps (~10.4% and ~6.6%).
 - Higher replay failure concentration (0.16) than no replay (0).
 - Good archive diversity (0.24), lower archive hardness.
 - Positive adaptation efficiency (0.0377).
- **Failure Replay** (phase6_failure_replay and phase6_residual_failure_replay):
 - Worse transfer and TSPLIB gaps (~16%+).
 - Highest replay failure concentration (0.29-0.33).
 - Large size bias (0.12).
 - Negative adaptation efficiency (~-0.03).
 - Lower archive descriptor diversity (~0.17).
- **Compression-aware variants** do not improve transfer gaps relative to their base counterparts, indicating compression reduces code novelty without proportional transfer improvement.
- **Stratified Random Replay** improves held-out gaps moderately but with no synthetic advantage, indicating selective sampling helps slightly.

Code Novelty vs. Transfer

- The condition phase6_diversity_failure_replay_compression shows **zero** last and mean code novelty but does **not** improve transfer beyond failure replay baselines.

- Other conditions with high mean code novelty (~0.8-0.9) do not necessarily translate to better transfer (e.g., failure replay with high novelty but worse transfer).
- Hence, **code novelty falls when transfer improves is not observed in this dataset**. Instead, high novelty alone does not guarantee better transfer.
- No clear lexical novelty to algorithmic invention mapping; improvement linked more to replay strategy and archive properties.

Detailed Replay Modes

- **Broad-Coverage Replay**: Not explicitly named, possibly akin to random replay; supported by higher archive diversity and lower failure concentration.
- **Raw-Failure Replay**: phase6_failure_replay and phase6_residual_failure_replay replicate failure cases, leading to high failure concentration, larger archive size bias, and poorer transfer.
- **Residual-Failure Replay**: Focus on remaining failure cases after prior handling; shows no transfer improvement over raw failure replay.
- **Diversity-Weighted Replay**: Emphasizes diverse failure cases with weights; yields strong archive diversity preserved but shows poor transfer similar to failure replays, possibly due to size bias.
- **Compression-Aware Replay**: Incorporates compression pressure; reduces code novelty significantly without transfer gains.

Failure Concentration and Transfer

- Conditions with **lower failure concentration** (random, diversity_failure) have superior transfer metrics.
- High failure concentration correlates with worse transfer and TSPLIB gaps.
- Selection diversity is higher in random replay (~0.26) and diversity_failure (~0.20), lower in failure replay (~0.17).
- Residual and failure replays show failure concentration ~0.3 with negative adaptation efficiency.

Training Gap and Worst Instances

- Training gaps mostly higher (~25-36%), no condition drastically reduces training gap to match synthetic holdout.
- Worst training instances remain consistent in difficulty across conditions.
- This supports that replay conditions improve generalization on held-out but not fully on training hard cases.

Conclusion

- **Phase6 diversity_failure_replay** is the best overall condition in terms of **transfer to TSPLIB and synthetic holdout sets**, with lowest optimality gaps in transfer metrics.
- **Random replay** yields best synthetic holdout zero gaps and solid transfer results.
- **Failure-based replay strategies**, including residual failure and diversity weighted replays with failure concentration and size bias, show **worse transfer and adaptation efficiency**.
- **Compression-aware replay reduces code novelty without transfer improvement**, suggesting compression should be used cautiously.
- Code novelty does not strongly correlate with transfer improvements; algorithmic gains are better identified by replay strategy effectiveness rather than lexical novelty alone.
- Replay archive properties (diversity, hardness, failure concentration, size bias) strongly influence transfer outcomes; low failure concentration and balanced archive hardness/diversity are favorable.
- Prioritize replay modes emphasizing diversity with failure cases but controlling failure concentration for improved TSP performance generalization.

Interpretation is restricted to objective metrics; no narrative speculation provided.